BEING A LOCAL COMPLETE INTERSECTION MORPHISM IS INHERITED FROM THE COMPOSITE OVER A SMOOTH BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

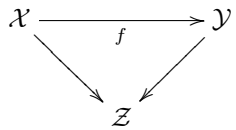
Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The definition of a local complete intersection morphism for algebraic stacks through smooth covers by schemes. The author's section on local complete intersection morphisms, in which the counted statement occurs. Reduces the assertion to the corresponding assertion for morphisms of schemes.
- The corresponding permanence statement for morphisms of schemes, together with the smoothness of the base change of the given smooth morphism. Both are used inside the counted proof (source lines 9949-9957), the first as a cross-chapter pointer. Gives the conclusion for the covering morphisms of schemes.

Source reference. Original reference more-morphisms-lemma-lci-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9956–9956. Navigation pointer only: the counted text refers to the Stacks item more-morphisms-lemma-lci-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let*



be a commutative diagram of morphisms of algebraic stacks. Assume $\mathcal{Y} \to \mathcal{Z}$ is smooth and $\mathcal{X} \to \mathcal{Z}$ is a local complete intersection morphism. Then $f : \mathcal{X} \to \mathcal{Y}$ is a local complete intersection morphism.

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Proof. Choose a scheme W and a surjective smooth morphism $W \to \mathcal{Z}$. Choose a scheme V and a surjective smooth morphism $V \to W \times_{\mathcal{Z}} \mathcal{Y}$. Choose a scheme U and a surjective smooth morphism $U \to V \times_{\mathcal{Y}} \mathcal{X}$. Then $U \to W$ is a local complete intersection morphism of schemes and $V \to W$ is a smooth morphism of schemes. By the result for schemes (More on Morphisms, Lemma more-morphisms-lemma-lci-permanence) we conclude that $U \to V$ is a local complete intersection morphism. By definition this means that f is a local complete intersection morphism. $\square$